

Chapter 151

Using Open FX and Resolve FX

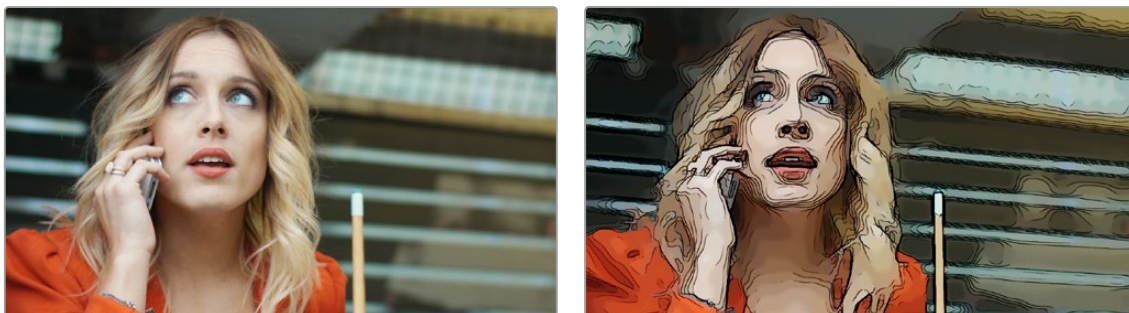
This chapter covers the use of Resolve FX and Open FX plugins, that allow you to use the built-in filters that come with DaVinci Resolve, as well as third-party filters from a variety of companies, to create complex effects and adjustments that aren't possible using the ordinary palette tools in the Color page.

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Resolve FX

Resolve FX are the built-in plugins that come with DaVinci Resolve. These plugins span the gamut from blurs and complex color adjustments to stylized image treatments and lighting effects to sharpen and repair operations that are too complex to accomplish using the palette controls of the Color page.



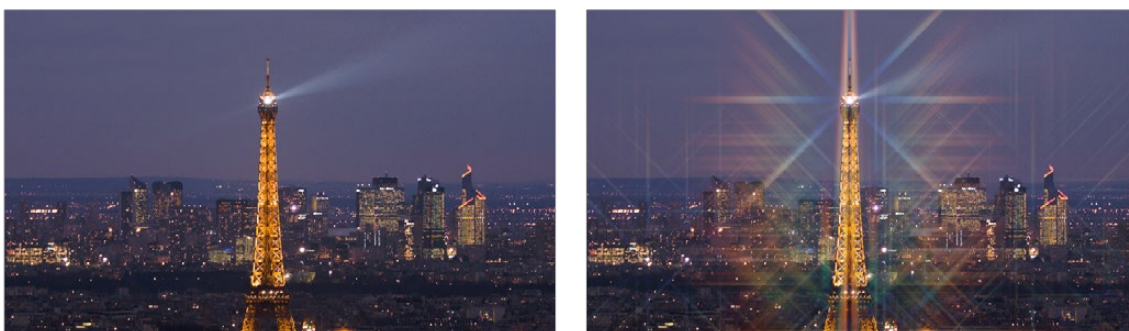
An image before/after using the Abstraction Resolve FX filter

Most Resolve FX plugins have been optimized for real-time playback, making it possible to apply complex effects such as Lens Flares, Light Rays, Film Grain, or Warping, and make adjustments while getting immediate, high-quality feedback, and enabling you to play each variation of your effect as you work without the need to wait for rendering or caching to happen first. Of course, if you're working with extremely high-resolution or raw source media, if your workstation is particularly old, or if you're applying many Resolve FX all at once, your performance may slow, necessitating the use of either the Smart Cache or User Cache.

Open FX

Open FX (OFX) is an open plugin standard intended to enable easier development of cross-platform visual effects plugins for a variety of applications. Popular plugin packages include BorisFX Sapphire and Continuum Complete, Red Giant Universe, and NewBlue TotalFX, all of which are ubiquitous tools for feature and broadcast work. The available Open FX plugin packages are also growing every year as this format becomes more widely adopted among developers.

With Open FX support, you can use plugins to do many stylized operations that would be difficult or impossible to do using the other tools in DaVinci Resolve. Everything from lens flares, optical blurs and prism effects, lens warp correction, film and video grain and damage effects, dead-pixel corrections, and more can be accomplished with the right plugin collection.



One of the many Sapphire OFX plugins from GenArts

The installation and licensing of Open FX plugins is handled by a vendor's own installer. Once installed, Open FX plugins appear within the Library of the Open FX panel, which can be opened by clicking the FX button at the top right of the Color page or Edit page Interface toolbar.

Where are OFX Installed?

According to the standard governing how OFX work, all OFX plugins on a particular workstation are installed into a standardized location to foster plugin compatibility with multiple applications. These locations are:

On macOS: /Library/OFX/Plugins

On Windows: C:/Program Files/Common Files/OFX/Plugins

On Linux: /usr/OFX/Plugins

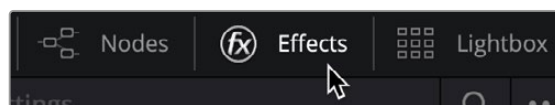
Open FX Plugins Can Be Processor Intensive

Because they create such a wide variety of effects, some third-party Open FX plugins can be extremely processor intensive, all the more if you add multiple plugins to a single grade. If you find your playback performance dropping because of a particularly expensive effects operation, you can use the Smart Cache to automatically cache nodes and clips that have Open FX plugins applied to them. Once fully cached, you can play these clips back in real time, at least until you change that clip's grade again.

For more information on caching and on improving performance in DaVinci Resolve overall, see *Chapter 8, "Improving Performance, Proxies, and the Render Cache."*

Browsing the Effects Library

All of these built-in plugins appear within categories at the top of the Effects Library.



Click the Effects button to open the Effects Library

When you click the Effects button, the Effects panel opens out of the right side of the Node Editor to show the Library, resizing the Viewer, Gallery, and Node Editor to make room. The Effects Library is organized hierarchically. Each vendor's plugins appear under a header with the name of that plugin collection, and possibly organized into categories, separated by headers with Open or Close arrows that appear to the right of the category name, which let you show or hide the contents so that you can make the hierarchy as compact or spread out as you like.

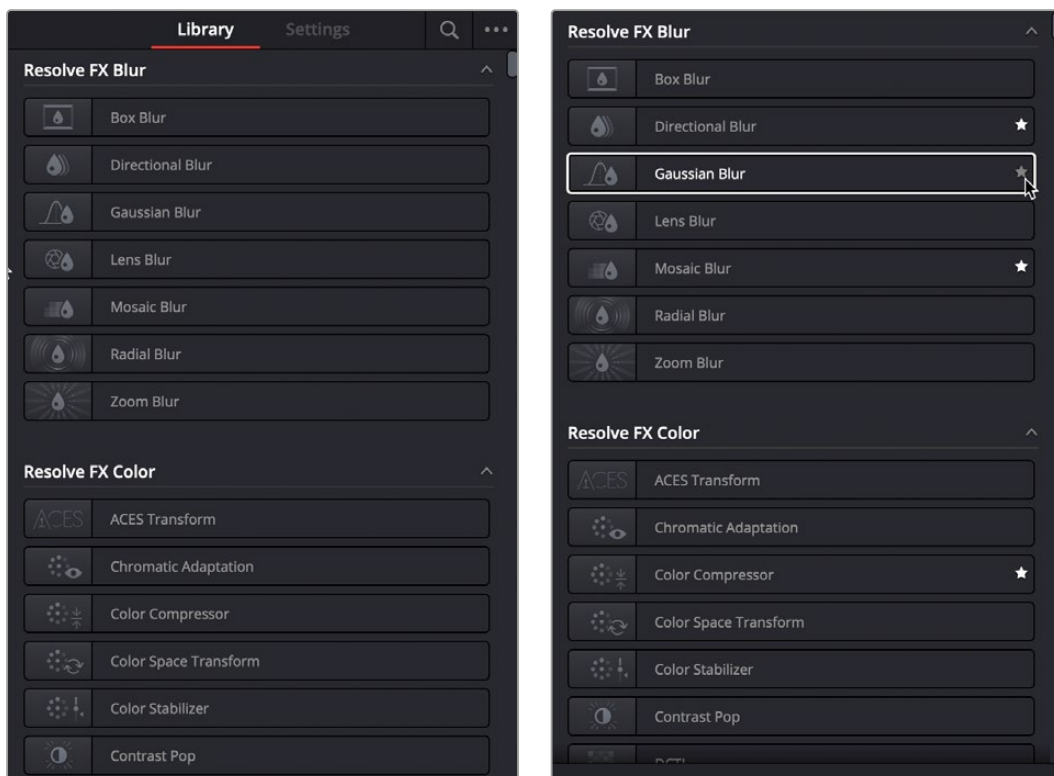
To open and close Effects categories, do one of the following:

- Move the pointer over the header you want to open or close, and click the Close or Open arrow to the right of the category name.
- To open or close all headers at once, Option-click the Open or Close arrow.

Since many Open FX plugin collections are quite large, an optional Search field can be opened at the top of the Library that lets you quickly find plugins by name or partial name.

To search for an Effects filter by name:

- 1 Click the magnifying glass button at the upper right-hand corner of the Effects panel.
- 2 Type your search string into the Search field that appears. A few letters should be enough to isolate only those plugins that have that character string within their name.



The Library in the Effects panel showing available effects

Stars indicate a flagged favorite effect

Effects Library Favorites

You can click on the far right of any Resolve FX or OFX filter to flag it with a star as a favorite filter. When you do so, choosing Favorites from the Effects Library option menu filters out all clips that are not favorites, letting you see only effects you most commonly use. To “de-favorite” any effect, click its star to turn it off.

Using Resolve FX and Open FX in the Color Page

This section provides an overview of procedures that describe how you can work with Open FX plugins within the Color page.

Methods of working with Open FX:

- **To add an Open FX plugin to a node:** Drag a plugin from the Open FX Library onto a node. If you drag a plugin onto a node that already has a plugin, the previous plugin will be overwritten.
- **To remove an Open FX plugin from a node:** There are two ways to delete an Open FX plugin from a node.

Right-click a node showing the FX badge, and choose Remove OFX Plugin from the contextual menu.

Click on the trashcan icon to the right of the name of the effect in the Effects Inspector.

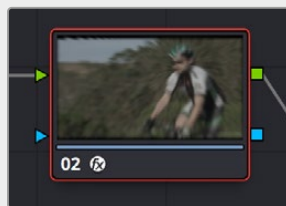
- **To edit the parameters of an Open FX plugin:** Select any node with an FX badge, and open the Open FX Panel to show the Settings list. You can switch to the Library by clicking the Library button.
- **To sample a color parameter within an Open FX plugin:** Some OFX plugins have a color swatch parameter, which exposes an eyedropper button in the Settings. Clicking the eyedropper turns the pointer into an eyedropper you can use to sample the contents of the Viewer.
- **To use Open FX onscreen controls in the Viewer:** Select any node with an FX badge, and the onscreen controls, if there are any, should appear in the Viewer. If not, make sure the Viewer mode drop-down is set to FX.

Applying Resolve FX and Open FX Plugins

Once you've found an Open FX plugin you want to use in the Library, there are two ways of applying it within the Node Editor of the Color page. Which method you use depends on how you want to use that plugin.

Adding a Plugin to an Existing Corrector Node

If you want to combine an Open FX plugin with a grade within a single node, simply drag and drop it onto a new corrector node to apply that plugin's effect to that node. Nodes with an Open FX plugin applied have an FX badge in the bottom left-hand corner.



A node with an
Open FX plugin mapped

You can only apply one Open FX plugin to a node at a time, but by using multiple nodes you can add as many Open FX plugins to your grade as you need.

When added to a corrector node, Open FX are applied after Motion Blur and Noise Reduction, but before anything else. This means you can use Motion Blur and Noise Reduction to pre-process the image before it's handed off to the Open FX plugin. This also means that all other adjustments you make within that node are applied to the Open FX plugin's output.

However, the principal advantage of adding plugins to corrector nodes is that you can use secondary operations such as a window, a qualifier, or a key to limit that plugin's effect, much as you would limit any other kind of adjustment you'd make with a corrector node.

Adding a Plugin as a New Corrector Node

If you want to add an Open FX plugin to your grade as a stand-alone effect, you can simply drag any plugin from the Open FX Library into the Node Editor or onto a connection line of your grade, and a new Corrector Node with the Open FX plugin applied will be created.

This new node will now include all the additional inputs and Alpha connectors needed to use the effect. So now that color correction nodes can be used for everything, there is no need to make separate FX nodes anymore. This should help simplify node trees and involve much less manual management of connections between nodes.

Adding a Plugin as a Stand-Alone FX Node (Legacy)

As part of an overhaul of the Resolve FX system to make it more intuitive and easier to use, dragging an effect from the Effects Library into the Node Editor will now create that effect using a standard corrector node and not an FX Node.

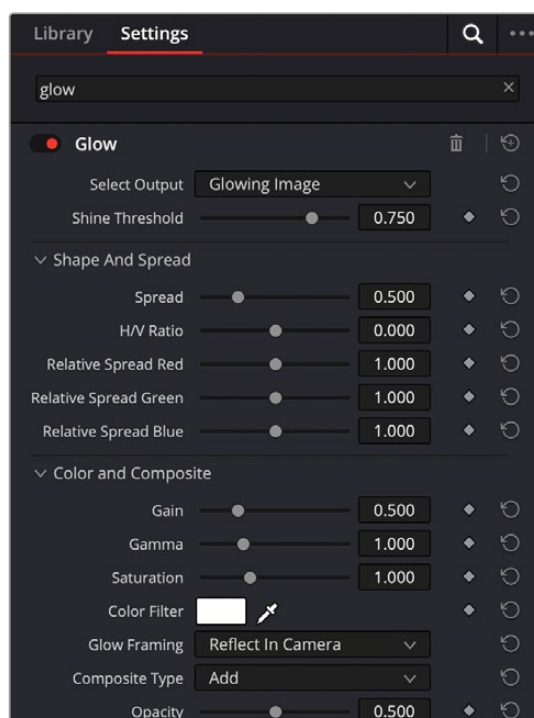
As of DaVinci Resolve 18.5, FX nodes have been depreciated. There is no need to make an FX node anymore as all of the previous functionality found in FX nodes have been rolled into the standard corrector node instead.

However for legacy projects, or if you still prefer to work in the old way, you can still force an FX node by option-dragging the effect from the Effects Library into the Node Editor instead of dragging it.

Resolve FX and Open FX Settings

When you select a node with a Resolve FX or Open FX plugin applied to it, the Open FX panel switches to the Settings, which show you every single parameter associated with that plugin, ready for customization.

Adjust any of the standard controls to manipulate that plugin's effect on the image.



The parameters of the Glow Resolve FX

Editing Effects Using the Full Screen Viewer

Because the Open FX panel can often be too short to present the full controls of more complicated filters, it remains visible when you switch to the Full Screen Viewer mode in the Color page. Jump into this mode by choosing Workspace > Viewer Mode > Full Screen Viewer (Shift-F).

This control layout makes it considerably easier to do detailed work while viewing a larger image and having all of your effects visible in a taller panel off to the side.

TIP: You can also open and close the Node Editor while in Full Screen Viewer mode, if you need to switch nodes while doing effects work.



The Full Screen Viewer mode of the Color page lets you keep the Settings panel open as you work

Resolve FX and Open FX Onscreen Controls

In the Edit page, Fusion page, and Color page, Resolve FX and Open FX display on-screen controls that you can use to visually edit an effect. In the Edit and Color pages, selecting an Open FX plugin node or a plugin in the Inspector that has onscreen controls automatically changes the Viewer's mode to Open FX Overlay mode, with the available controls ready to use. Different plugins expose different custom controls, letting you control the effect or manipulate the image, depending on that plugin's function.



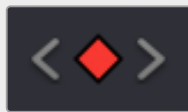
Adjusting the onscreen controls exposed by the GenArts Sapphire Glint Rainbow plugin

If for whatever reason you switch the Viewer to another onscreen control mode (for example, showing the Window or Image Wipe controls), you can always switch back to the Open FX controls by choosing the Open FX Overlay mode from the onscreen control drop-down menu underneath the Viewer.

TIP: You may find that as you work you want to temporarily hide or show the onscreen controls in the Viewer so you can get an uncluttered look at the image you're adjusting. You can quickly toggle any set of onscreen controls off and on without selecting Off in the menu by pressing Shift-` (tilde).

Keyframing Resolve FX and OFX in the Inspector

Resolve FX and Open FX can be keyframed in the Edit, Fusion, and Color pages. However, they can only be keyframed in the Edit and Color pages using the keyframing controls found in the Inspector (at the time of this writing). Happily, most simple keyframing tasks can be performed using three buttons that appear to the right of any parameter that's capable of being keyframed. It takes two keyframes at minimum to create an animated effect.



The three keyframe controls that appear in the Inspector, from left to right: Previous keyframe, Create/Delete keyframe, Next keyframe

Methods of keyframing parameters in the Inspector:

- **To add a keyframe:** Select a clip, open the Inspector, then move the Timeline playhead to the frame where you want to place a keyframe, and click the Keyframe button next to the parameter of the Inspector you want to animate. Once you've added at least one keyframe to a parameter, all other adjustments you make to parameters in the Inspector, or using the onscreen Transform/Crop controls in the Timeline Viewer, add new keyframes automatically if the playhead is at another frame.
- **To move the playhead to the next or previous keyframe:** Click the small left- or right-hand arrow to either side of a parameter's keyframe control, or press Right-Bracket (]) or Left-Bracket ([), to jump the playhead to the next or previous keyframe.
- **To edit an existing keyframe of a parameter:** Move the playhead to be on top of the keyframe you want to edit, and then change that parameter, either in the Inspector, or using the onscreen controls of the Timeline Viewer.

Methods of changing keyframe interpolation in the Inspector:

- **To change a keyframe to Static:** (Color page only) Static keyframes create abrupt one-frame changes at the keyframe to which they're applied, which is good for creating sudden effects. Move the playhead to a frame with a keyframe using the next/previous keyframe controls, then right-click the orange keyframe button and choose "Change to Static Keyframe." The keyframe control changes to a round button to show that keyframe is now Static.
- **To change a keyframe to Dynamic:** Move the playhead to a frame with a keyframe using the next/previous keyframe controls, then right-click the orange keyframe button and choose "Change to Dynamic Keyframe." The keyframe control changes to a diamond button to show that keyframe is now Dynamic.

Methods of deleting keyframes and disabling keyframed effects:

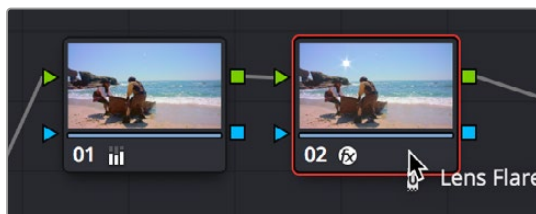
- **To delete a single keyframe:** Open the Inspector, move the Timeline playhead to a frame with a keyframe, and click the orange Keyframe button in the Inspector to delete it.
- **To delete all keyframes for one parameter:** Click the reset button to the right of a parameter's keyframe control in the Inspector.
- **To delete all keyframes in a group of parameters in the Inspector:** Click the reset button to the right of a parameter group's title bar in the Inspector.
- **To disable or enable a single parameter's keyframed effect:** In the Timeline, click the toggle control at the left of a parameter's keyframe track. A white dot means it's enabled, while no dot means it's grayed-out and disabled.
- **To disable or enable a group of parameters in the Inspector:** Click the toggle control at the left of a parameter group's title bar in the Inspector. Orange means that group is enabled. Gray is disabled.

Motion Tracking Resolve FX and Compatible OFX Plugins

When using Resolve FX in the Color page, Resolve FX that have position parameters, including Dent, Lens Flare, Light Rays (when "From a Location" is selected), Mirrors, Radial Blur, Ripples, Vortex, and Zoom Blur can all be motion tracked to follow the position of a moving subject in the frame using the easier and more reliable IntelliTrack mode (described below) or the legacy point-based tracking in the FX mode of the Tracker palette.

To match move Resolve FX to a feature using the FX tracker:

- 1 Create a new node, and drag the Resolve FX filter you want to apply onto that node to apply the effect. In this example, we're adding a Lens Flare effect, that has the position parameters necessary to be match moved to follow the motion of the shot, and we're choosing the MIR-I 2.8/37 preset, which looks like a sun.



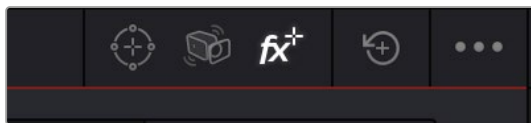
Applying a Lens Flare to the shot

- 2 If necessary, use the onscreen controls in the Viewer or the X Position and Y Position sliders to move the Resolve FX effect to where you want it.



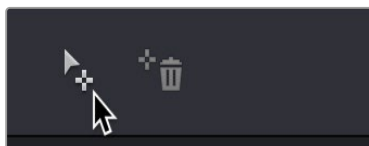
Moving the Lens Flare using the onscreen controls

- 3 Next, open the Tracker Palette and choose the fx icon in the upper right.



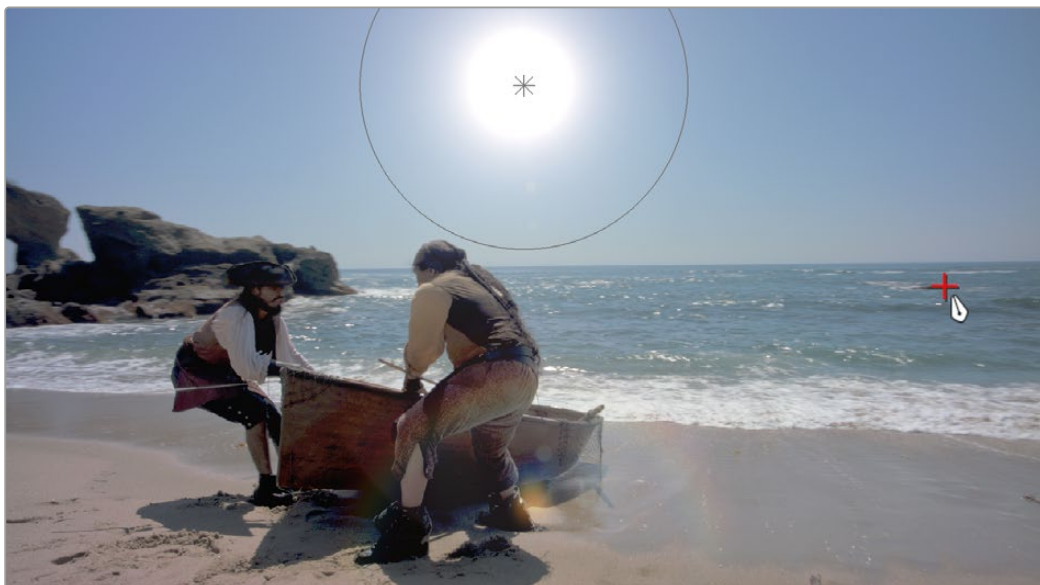
Opening the FX tracker mode of the Tracker palette

- 4 Click the Add Tracker Point button, at the bottom left-hand side of the Tracker palette, to add tracker crosshairs to the center of the Viewer.



Adding a point tracker

- 5 Drag the crosshairs to a high-contrast detail (such as a small object or corner), and click the Track Forward button. In this example, there's a rock out at sea that will make a good plane of motion for tracking a far-away sun. There is no inner or outer box to position or resize while you do this; you just need to drag the crosshairs to center on the feature you want to track.



Positioning the point tracker on a feature you want to track

- 6 Now, click the Track Forward button, and DaVinci Resolve will track the feature. The Resolve FX applied to that node will simultaneously move to follow the track, and when the tracking is done, you're finished.



A successful track

- 7 After you've finished tracking, you can freely reposition the Resolve FX to offset it from the track.

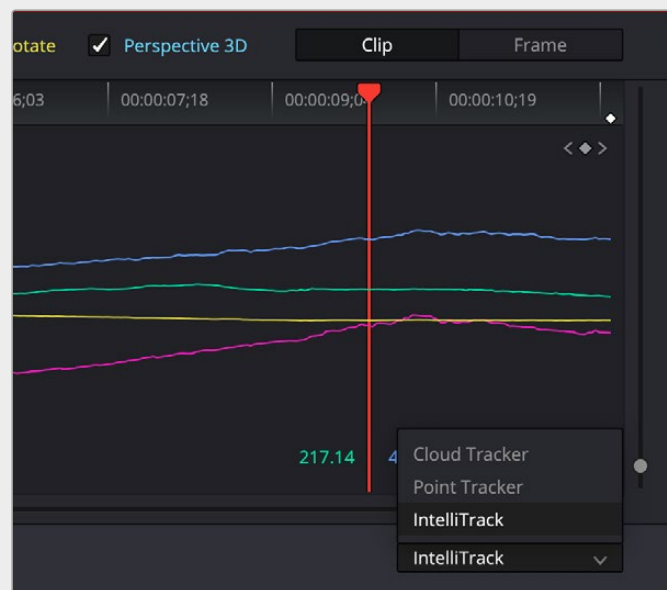
For more information about single-point tracking, refer to the "Point Tracker Workflows" section, see *Chapter 140, "Motion Tracking Windows."*

IntelliTrack AI Point Tracker (Studio Version Only)

The DaVinci Neural Engine powered IntelliTrack point tracker is available in the FX tracker in DaVinci Resolve Studio. The IntelliTrack setting can be used for tracking your Resolve FX.

IntelliTrack, like other AI tools, doesn't use predetermined rulesets or heuristics. Instead, it is trained on real world examples. This makes it less likely to fail in scenarios like tracking a subject behind brief occlusions. For most cases it will be more precise and more robust than the normal Point Tracker.

Using IntelliTrack: All the IntelliTrack analysis happens behind the scenes, and there is no additional user input required other than to choose the IntelliTrack setting, set the tracking crosshairs over high contrast areas on the same moving subject, and then press one of the tracker direction controls.



Setting the IntelliTrack mode for the Color page window tracker